EUCLIDEAN LOBE GEOMETRY, RECIPROCAL SCATTERING, AND TRANSVERSE PROJECTIVE STRUCTURE OF THE RIEMANN ZETA FUNCTION

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Abstract. The curve

$$\Gamma(t) = \left(t, \operatorname{Re}\zeta\left(\tfrac{1}{2}+it\right), \operatorname{Im}\zeta\left(\tfrac{1}{2}+it\right)\right) \tag{0.1}$$

is treated as a primary Euclidean object and decomposed into zero-to-zero lobes. The construction combines radial excursion, angular sweep, Euclidean arc length, eccentric lobe geometry, reciprocal scattering, and the exact transverse equation

$$F_\delta = -iF_t, \qquad F(\delta,t) = \zeta\left(\tfrac{1}{2}+\delta+it\right). \tag{0.2}$$

The Riemann–von Mangoldt/Lambert phase produces an exact tangent-circle model whose reciprocal image is a straight scattering branch and whose transverse continuation satisfies an explicit no-exit theorem. For the exact zeta lobes we derive a reciprocal normal form, a corrected endpoint scattering cocycle, and a projective defect based on the Schwarzian derivative. The sign and reciprocal conventions in the endpoint jet formulas are fixed explicitly: the relevant quotient is $w_n'(1)/w_n'(0)$, not its reciprocal, and the cocycle potential is $Q_n = -\log a_n - i\psi_n$.

The projective block yields an exact second-order equation, an exact closure orthogonality, an exact complex moment formula for the endpoint jet holonomy, and a telescoping product law. These identities are structural rather than a confinement proof: no independent inequality forcing zero-exclusion is obtained from the Schwarzian alone. The reciprocal chart $U = 1/F$ is also treated with its exact logical status: a pole of U and a zero of F are the same event at the same domain point. Its utility is methodological, not logical, because a lower-bound problem for $|F|$ becomes an upper-bound problem for $|U|$ only on regions where the reciprocal chart is already holomorphic.

The numerical figures and tables carried into this version are unchanged from the preceding verified computation. Version 5 is an analytic and typographical revision; the binary corpus was not re-inspected or recomputed.

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1. Status and scope

The purpose of this paper is to retain one geometric object throughout: the trace in (0.1), referred to as the *Violet Thread*. The main statements are separated by mathematical status.

Proposition 1.1 (Exact structural layer)**.** *The following statements are exact consequences of the definitions used below: the zero-to-zero lobe decomposition, the Hardy-frame speed identity, the transverse quarter-turn equation* (0.2)*, the RVM tangent-circle identity, the RVM reciprocal straight-line identity, the RVM no-exit theorem, the reciprocal curvature formula, the corrected scattering cocycle, the endpoint-normalized lobe factorization, and the Schwarzian closure and holonomy identities.*

Remark 1.2 (Logical status of transverse confinement). The assertion

$$F(\delta,t)\neq 0 \quad \text{for every } t\in\mathbb{R} \text{ and } 0<|\delta|<\tfrac{1}{2} \tag{1.1}$$

is exactly the desired global zero-confinement statement. Equation (0.2) is a complete first-order evolution equation, so a proof of (1.1) from a global invariant of that first-order field would require no subsequent hierarchy of second-, third-, or higher-order conditions. Equation (0.2) by itself does not prove (1.1).

Remark 1.3 (Numerical layer). The reported correlations for the first 49 lobes and the full 2,001,052-ordinate phase/RVM summaries are carried unchanged from the preceding verified computation. They are not recomputed in Version 5.

2. Notation

Symbol	Meaning
$W(t)$	$\zeta(\frac{1}{2}+it)$, the complex value trace of the Violet Thread
$\Gamma(t)$	$(t,\operatorname{Re}W(t),\operatorname{Im}W(t))\in\mathbb{R}^3$
$Z(t)$	Hardy's real function, $W(t)=Z(t)e^{-i\vartheta(t)}$
$\vartheta(t)$	Riemann–Siegel phase
γ_n	successive zeros of W on the central trace
I_n	(γ_n,γ_{n+1})
g_n	$\gamma_{n+1}-\gamma_n$
Ω_n	$\vartheta(\gamma_{n+1})-\vartheta(\gamma_n)$
s_n	Ω_n/π
A_n	$\max_{I_n}\lvert Z(t)\rvert$
L_n	Euclidean arc length of the nth lobe
R_n	L_n/g_n
$F(\delta,t)$	$\zeta(\frac{1}{2}+\delta+it)$
$U(\delta,t)$	reciprocal chart $1/F(\delta,t)$ where defined
Θ_0	smooth RVM phase $\frac{z}{2}\log(z/(2\pi e))-\frac{\pi}{8}$
τ_n	RVM/Lambert ordinates
$\mathcal{C}_n$	phase centroid of a rotated lobe
E_n	normalized eccentricity $2\lvert\mathcal{C}_n\rvert/A_n$
D_n	normalized circle-tangency defect
$\mathcal{H}_n$	endpoint jet quotient $w_n'(1)/w_n'(0)$
Q_n	corrected scattering potential $-\log a_n-i\psi_n$
$\mathcal{P}_n$	projective lobe defect $\{w_n,q\}-2\pi^2$

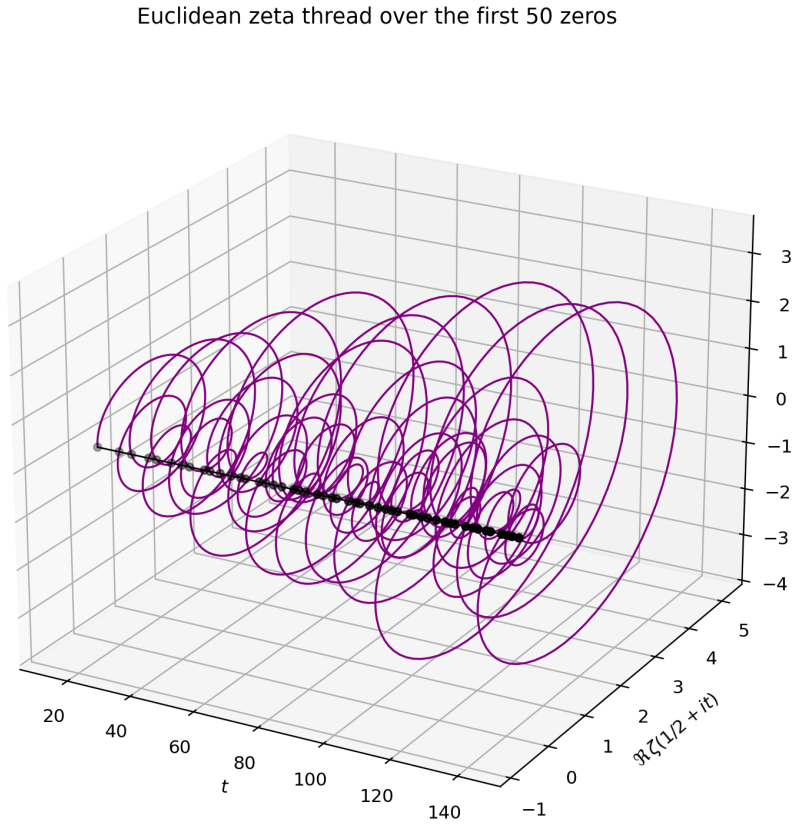


FIGURE 1. The Euclidean Violet Thread over the first fifty carried zero ordinates. Height remains an explicit coordinate, so the transverse lobes are genuine three-dimensional curve segments.

3. The Violet Thread and its zero-to-zero lobes

Definition 3.1 (Violet Thread)**.** Define

$$W(t) = \zeta\left(\frac{1}{2} + it\right) = u(t) + iv(t), \tag{3.1}$$

and

$$\Gamma(t) = \big(t, u(t), v(t)\big). \tag{3.2}$$

The classical Hardy decomposition [1, 2] is

$$W(t) = Z(t)e^{-i\vartheta(t)}, \qquad Z(t) \in \mathbb{R}. \tag{3.3}$$

Consequently

$$\Gamma(t) = \big(t, Z(t)\cos\vartheta(t), -Z(t)\sin\vartheta(t)\big). \tag{3.4}$$

Differentiation gives

$$W_t(t) = e^{-i\vartheta(t)}\big(Z'(t) - i\vartheta'(t)Z(t)\big) \tag{3.5}$$

and hence

$$|W_t(t)|^2 = Z'(t)^2 + \vartheta'(t)^2 Z(t)^2. \tag{3.6}$$

Let $\gamma_n < \gamma_{n+1}$ be consecutive contacts of the central trace with the value-plane origin. On

$$I_n = (\gamma_n, \gamma_{n+1}), \tag{3.7}$$

the sign of Z is constant. Put

$$\varepsilon_n = \operatorname{sgn} Z(t), \qquad t \in I_n, \tag{3.8}$$

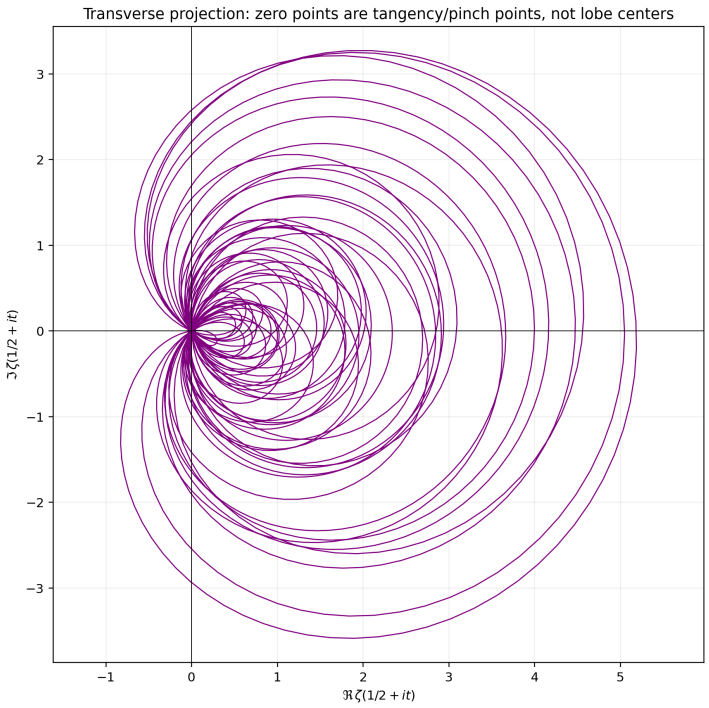


FIGURE 2. Transverse projection. The common zero point is an eccentric contact of the lobes, not their geometric center.

and rotate away the absolute initial phase:

$$\widehat{W}_n(t) = \varepsilon_n e^{i\vartheta(\gamma_n)} W(t). \tag{3.9}$$

Then

$$\widehat{W}_n(t) = |Z(t)| e^{-i\phi_n(t)}, \qquad \phi_n(t) = \vartheta(t) - \vartheta(\gamma_n). \tag{3.10}$$

The total angular sweep is

$$\Omega_n = \vartheta(\gamma_{n+1}) - \vartheta(\gamma_n), \qquad s_n = \frac{\Omega_n}{\pi}. \tag{3.11}$$

Thus the usual local phase unfolding is literally the angular extent of a single lobe.

4. Eccentric one-arch geometry and the RVM skeleton

Introduce uniform phase coordinate

$$q = \frac{\phi_n(t)}{\Omega_n}, \qquad 0 \le q \le 1. \tag{4.1}$$

The one-arch radial model retaining the measured phase sweep is

$$\widehat{W}_n^{\mathrm{arch}}(q) = A_n \sin(\pi q) e^{-i\pi s_n q}. \tag{4.2}$$

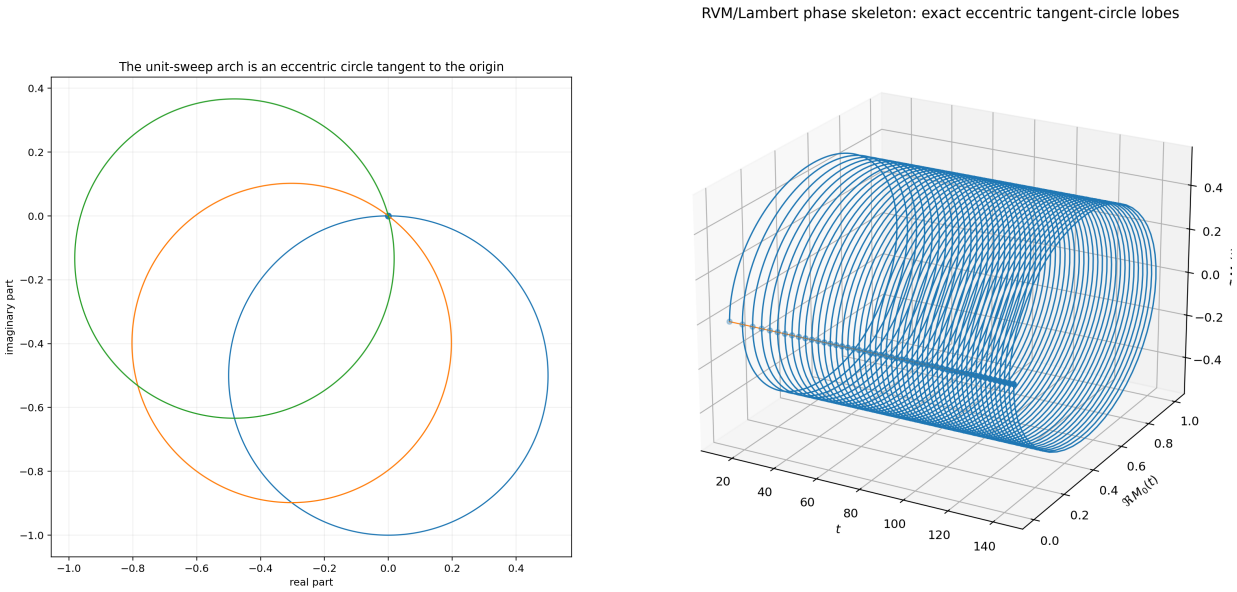
Theorem 4.1 (Unit sweep is an eccentric tangent circle)**.** *If $s_n = 1$, then*

$$\widehat{W}_n^{\mathrm{arch}}(q) = -\frac{iA_n}{2} + \frac{iA_n}{2} e^{-2\pi i q}, \tag{4.3}$$

and therefore

$$\left| \widehat{W}_n^{\mathrm{arch}}(q) + \frac{iA_n}{2} \right| = \frac{A_n}{2}. \tag{4.4}$$

The origin lies on the circle at a tangency/contact point and is not its center.



(A) Unit-sweep tangent-circle identity. (B) RVM/Lambert tangent-circle skeleton.

FIGURE 3. The eccentric geometry generated by a unit RVM phase sweep.

Proof. Use

$$\sin(\pi q)e^{-i\pi q} = \frac{1 - e^{-2\pi i q}}{2i}. \tag{4.5}$$

Substitution into (4.2) gives (4.3), and (4.4) follows. □

The smooth RVM phase is

$$\Theta_0(z) = \frac{z}{2}\log\frac{z}{2\pi e} - \frac{\pi}{8}. \tag{4.6}$$

The Lambert representation of the corresponding ordinates is standard [4]; with the normalization used here they satisfy

$$\Theta_0(\tau_n) = \pi\left(n - \frac{3}{2}\right). \tag{4.7}$$

Define the model thread

$$M_0(t) = \cos\Theta_0(t)e^{-i\Theta_0(t)}. \tag{4.8}$$

Then

$$M_0(t) = \frac{1}{2}\left(1 + e^{-2i\Theta_0(t)}\right), \tag{4.9}$$

so

$$\left|M_0(t) - \frac{1}{2}\right| = \frac{1}{2}. \tag{4.10}$$

Thus the RVM/Lambert skeleton consists of exact eccentric tangent-circle lobes.

5. Exact first-order transverse dynamics

Set

$$F(\delta, t) = \zeta\left(\frac{1}{2} + \delta + it\right), \qquad -\frac{1}{2} < \delta < \frac{1}{2}. \tag{5.1}$$

Theorem 5.1 (Global quarter-turn equation)**.** *The complete transverse family satisfies*

$$F_\delta(\delta, t) = -iF_t(\delta, t). \tag{5.2}$$

In the value plane the transverse and longitudinal velocities are orthogonal and have equal magnitude.

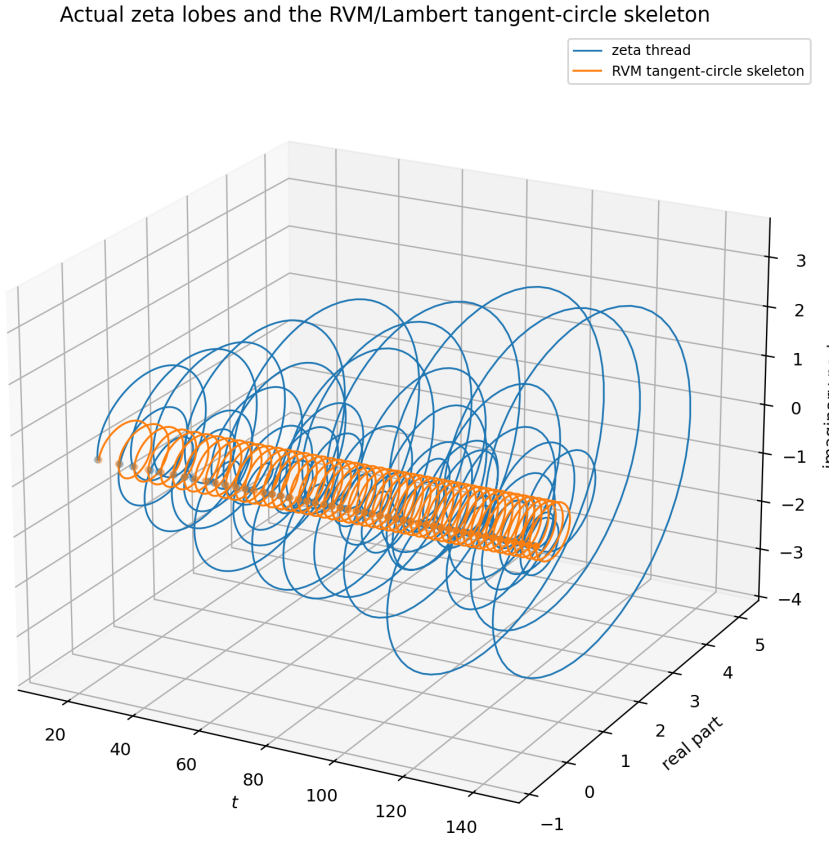


FIGURE 4. Actual zeta lobes and the RVM/Lambert tangent-circle skeleton in the same Euclidean coordinates.

Proof. If $s = \frac{1}{2} + \delta + it$, then

$$F_\delta = \zeta'(s), \qquad F_t = i\zeta'(s), \tag{5.3}$$

which gives (5.2). □

On the central thread, (3.5) gives

$$F_\delta(0,t) = e^{-i\vartheta(t)}\big(-\vartheta'(t)Z(t) - iZ'(t)\big). \tag{5.4}$$

Thus the Hardy frame separates the transverse generator into radial and angular components:

$$N_{\mathrm{rad}}(t) = -\vartheta'(t)Z(t), \qquad N_{\mathrm{ang}}(t) = -Z'(t). \tag{5.5}$$

Remark 5.2 (First order means the complete evolution law). Repeated differentiation of (5.2) yields identities such as

$$F_{\delta\delta} = -F_{tt}, \qquad F_{\delta\delta\delta} = iF_{ttt}, \tag{5.6}$$

but these are consequences of the same first-order equation and are not independent dynamical conditions.

6. Exact RVM no-exit theorem

Analytically continue the RVM skeleton by

$$M(\delta,t) = \frac{1}{2}\left(1 + e^{-2i\Theta_0(t-i\delta)}\right). \tag{6.1}$$

It satisfies the same quarter-turn equation

$$M_\delta = -iM_t. \tag{6.2}$$

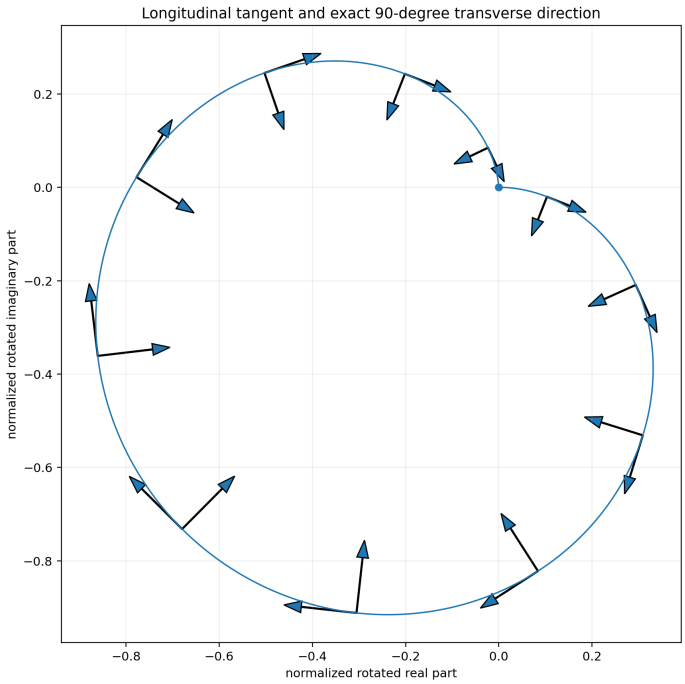


FIGURE 5. Longitudinal tangent vectors and their exact quarter-turned transverse directions along a representative lobe.

Theorem 6.1 (RVM tangent-circle no-exit)**.** *For $t > 2\pi$ and $\delta \neq 0$,*

$$M(\delta, t) \neq 0. \tag{6.3}$$

Moreover,

$$|M(\delta, t)| \geq \frac{1}{2}\left|1 - \left(\frac{t}{2\pi}\right)^{-\delta}\right| > 0. \tag{6.4}$$

Proof. Let

$$v(\delta) = \operatorname{Im} \Theta_0(t - i\delta). \tag{6.5}$$

Since

$$\Theta_0'(z) = \frac{1}{2}\log\frac{z}{2\pi}, \tag{6.6}$$

we obtain

$$v'(\delta) = -\frac{1}{2}\log\frac{\sqrt{t^2+\delta^2}}{2\pi}. \tag{6.7}$$

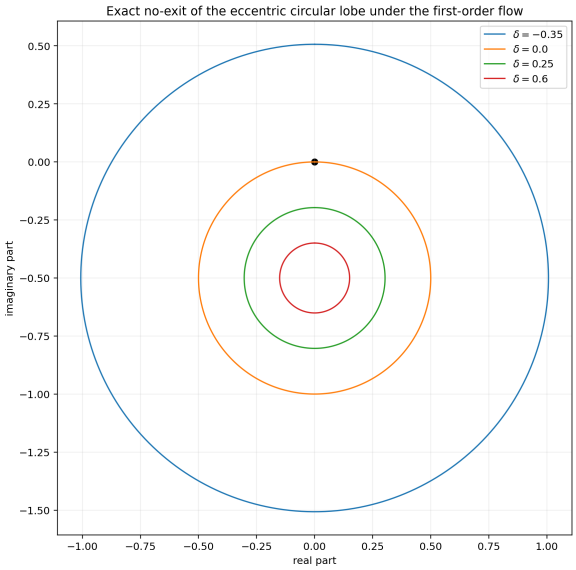
For $t > 2\pi$, v' is strictly negative and $v(0) = 0$, so $v(\delta)$ has the opposite sign to δ. If $M(\delta, t) = 0$, then $e^{-2i\Theta_0(t-i\delta)} = -1$, whose modulus is one. However,

$$\left|e^{-2i\Theta_0(t-i\delta)}\right| = e^{2v(\delta)} \neq 1 \tag{6.8}$$

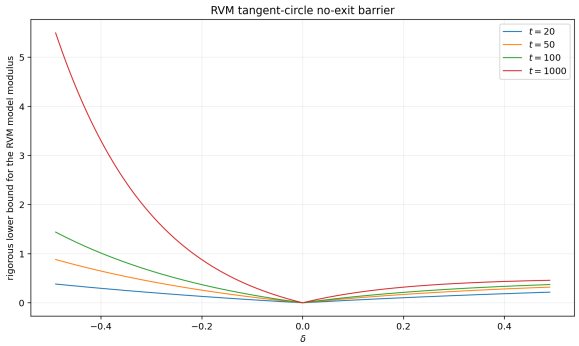
for $\delta \neq 0$, proving (6.3). Finally,

$$|v(\delta)| \geq \frac{|\delta|}{2}\log\frac{t}{2\pi} \tag{6.9}$$

and the reverse triangle inequality applied to (6.1) gives (6.4). □



(A) Constant-speed circular model.



(B) Explicit RVM modulus barrier.

FIGURE 6. Two representations of the exact first-order no-exit mechanism in the RVM tangent-circle model.

7. Reciprocal chart: exact equivalence and actual methodological gain

Where $F \neq 0$, define

$$U(\delta, t) = \frac{1}{F(\delta, t)}. \tag{7.1}$$

Proposition 7.1 (Zero–pole equivalence at the same domain point)**.** *Let $p = (\delta_0, t_0)$ and suppose F is analytic in the corresponding complex domain coordinate. If F has a zero of order m at p, then $U = 1/F$ has a pole of order m at the same point, and conversely:*

$$\operatorname{ord}_p(U) = -\operatorname{ord}_p(F). \tag{7.2}$$

Thus the statement "a new pole appears in U" is exactly the statement "a new zero appears in F" viewed in the reciprocal chart of the Riemann sphere.

Proof. Write

$$F(z) = (z - z_0)^m h(z), \qquad h(z_0) \neq 0. \tag{7.3}$$

Then

$$U(z) = (z - z_0)^{-m} h(z)^{-1}, \tag{7.4}$$

which proves (7.2). □

Proposition 7.2 (Quarter-turn invariance under inversion)**.** *On every domain where U is defined,*

$$U_\delta = -iU_t. \tag{7.5}$$

Proof. From (7.1) and (5.2),

$$U_\delta = -\frac{F_\delta}{F^2} \tag{7.6}$$

$$= \frac{iF_t}{F^2} = -iU_t. \tag{7.7}$$

□

Remark 7.3 (Methodological, not logical, change). On a region D where the reciprocal chart is holomorphic, the inequalities

$$\sup_D |U| \leq M \quad \Longleftrightarrow \quad \inf_D |F| \geq M^{-1} \tag{7.8}$$

are identical information. The reciprocal chart can nevertheless alter the available method because upper bounds for holomorphic functions can interact naturally with maximum-modulus and Phragmén–Lindelöf arguments. Establishing the holomorphy or pole-exclusion required to use those tools is part of the same zero-confinement problem; inversion does not create an independent RH criterion.

The RVM tangent circle becomes particularly simple under (7.1). For

$$\widehat{W}_{\mathrm{RVM}}(\phi) = A\sin\phi\, e^{-i\phi}, \qquad 0<\phi<\pi, \tag{7.9}$$

we have

$$U_{\mathrm{RVM}}(\phi) = \frac{1}{A}\cot\phi + \frac{i}{A}, \tag{7.10}$$

which is an exact straight line. The two zero contacts of the direct lobe become the two infinite ends of the reciprocal branch.

8. Reciprocal curvature and scattering normal form

Write a rotated lobe as

$$\widehat{W}_n(\phi) = r_n(\phi)e^{-i\phi}, \qquad 0<\phi<\Omega_n, \tag{8.1}$$

and define

$$U_n(\phi) = \frac{e^{i\phi}}{r_n(\phi)}. \tag{8.2}$$

Proposition 8.1 (Reciprocal curvature)**.** *The signed curvature of the reciprocal branch has the form*

$$\kappa_{U_n}(\phi) = \frac{r_n(\phi)^3\big(r_n(\phi)+r_n''(\phi)\big)}{\big(r_n(\phi)^2+r_n'(\phi)^2\big)^{3/2}}. \tag{8.3}$$

Hence

$$\operatorname{sgn}\kappa_{U_n} = \operatorname{sgn}\big(r_n+r_n''\big). \tag{8.4}$$

Proof. In polar form the reciprocal radius is $\rho = 1/r_n$. Substitution into the standard polar-curvature expression

$$\kappa = \frac{\rho^2+2\rho'^2-\rho\rho''}{(\rho^2+\rho'^2)^{3/2}} \tag{8.5}$$

gives (8.3). □

For the one-arch family $r(\phi) = A\sin(\phi/s_n)$,

$$r''(\phi)+r(\phi) = \left(1-\frac{1}{s_n^2}\right)r(\phi), \tag{8.6}$$

so the RVM value $s_n = 1$ is precisely the zero-curvature reciprocal state.

Now assume the two endpoint zeros of the lobe are simple and define

$$a_n = \frac{|Z'(\gamma_n)|}{\vartheta'(\gamma_n)}, \qquad a_{n+1} = \frac{|Z'(\gamma_{n+1})|}{\vartheta'(\gamma_{n+1})}. \tag{8.7}$$

The reciprocal asymptotics at the two ends show that the scattering-angle defect is

$$\Delta_n^{\mathrm{sc}} = \Omega_n - \pi = \pi(s_n-1). \tag{8.8}$$

In uniform phase coordinate $q = \phi/\Omega_n$, write

$$\widehat{W}_n(q) = \sin(\pi q)e^{-i\pi s_n q}H_n(q), \qquad H_n(q)>0. \tag{8.9}$$

The endpoint values are

$$H_n(0) = s_n a_n, \qquad H_n(1) = s_n a_{n+1}. \tag{8.10}$$

Define the endpoint-normalized interior residual

$$J_n(q) = \frac{H_n(q)}{H_n(0)^{1-q} H_n(1)^q}. \tag{8.11}$$

Then

$$J_n(0) = J_n(1) = 1, \qquad J_n(q) > 0 \quad (0 < q < 1), \tag{8.12}$$

and the exact lobe normal form is

$$\widehat{W}_n(q) = s_n \sin(\pi q) e^{-i\pi s_n q} a_n^{1-q} a_{n+1}^q J_n(q). \tag{8.13}$$

Proposition 8.2 (Integrated reciprocal-curvature budget)**.** *Let $\ell_n(q) = \log H_n(q)$ and define*

$$V_n(q) = \pi^2 s_n^2 \frac{r_{\phi\phi} + r}{r}. \tag{8.14}$$

Then

$$V_n = \ell_n'' + \ell_n'^2 + 2\pi \cot(\pi q) \ell_n' + \pi^2 (s_n^2 - 1), \tag{8.15}$$

and

$$\int_0^1 \sin^2(\pi q) V_n(q)\, \mathrm{d}q = \int_0^1 \sin^2(\pi q) \ell_n'(q)^2\, \mathrm{d}q + \frac{\pi^2}{2}(s_n^2 - 1). \tag{8.16}$$

Proof. Substitute $r(q) = \sin(\pi q) H_n(q)$ and $\phi = \pi s_n q$ into (8.14). Multiplication by $\sin^2(\pi q)$ turns the linear derivative terms into

$$\frac{\mathrm{d}}{\mathrm{d}q} \left(\sin^2(\pi q) \ell_n'(q) \right), \tag{8.17}$$

whose endpoint contribution vanishes, giving (8.16). □

9. Corrected endpoint scattering cocycle

The endpoint quotient must be fixed before any projective construction is used.

Proposition 9.1 (Correct endpoint jet quotient)**.** *For a sign-normalized lobe*

$$w_n(q) = r_n(q) e^{-i\pi s_n q} \tag{9.1}$$

with simple endpoint zeros,

$$w_n'(0) = \pi s_n a_n, \qquad w_n'(1) = -\pi s_n a_{n+1} e^{-i\pi s_n}. \tag{9.2}$$

Therefore

$$\mathcal{H}_n := \frac{w_n'(1)}{w_n'(0)} = \frac{a_{n+1}}{a_n} e^{-i(\Omega_n - \pi)}. \tag{9.3}$$

The reciprocal quotient $w_n'(0)/w_n'(1)$ is not the holonomy used below.

Proof. Equation (8.10) gives

$$r_n'(0) = \pi s_n a_n, \qquad r_n'(1) = -\pi s_n a_{n+1}. \tag{9.4}$$

At the endpoints the terms proportional to r_n vanish in the derivative of (9.1), yielding (9.2). Since $-e^{-i\pi s_n} = e^{-i\pi(s_n - 1)}$, equation (9.3) follows. □

Define

$$b_n = \log a_n, \qquad \psi_n = \vartheta(\gamma_n) - n\pi. \tag{9.5}$$

Then

$$\psi_{n+1} - \psi_n = \Omega_n - \pi. \tag{9.6}$$

The corrected complex potential is

$$Q_n = -b_n - i\psi_n. \tag{9.7}$$

Its conjugate satisfies

$$\overline{Q}_n - \overline{Q}_{n+1} = \log \frac{a_{n+1}}{a_n} - i(\Omega_n - \pi), \tag{9.8}$$

so

$$\mathcal{H}_n = e^{\overline{Q}_n - \overline{Q}_{n+1}}. \tag{9.9}$$

Equivalently, the oppositely oriented scattering coefficient

$$\mathfrak{S}_n = \frac{a_{n+1}}{a_n} e^{i(\Omega_n - \pi)} = e^{Q_n - Q_{n+1}} \tag{9.10}$$

satisfies $\mathcal{H}_n = \overline{\mathfrak{S}_n}$.

10. Projective lobe defect: exact identities and corrected branch

This section records a projective structure that is exact but is not promoted to a confinement criterion.

Definition 10.1 (Schwarzian projective defect)**.** For a lobe with $w_n'(q) \neq 0$, define

$$\{w_n, q\} = \frac{w_n'''(q)}{w_n'(q)} - \frac{3}{2}\left(\frac{w_n''(q)}{w_n'(q)}\right)^2 \tag{10.1}$$

and

$$\mathcal{P}_n(q) = \{w_n, q\} - 2\pi^2. \tag{10.2}$$

Lemma 10.2 (No interior critical point in the lobe parametrization)**.** *If $s_n > 0$ and $r_n(q) > 0$ for $0 < q < 1$, then*

$$w_n'(q) \neq 0, \qquad 0 < q < 1. \tag{10.3}$$

Proof. From (9.1),

$$w_n'(q) = e^{-i\pi s_n q}\left(r_n'(q) - i\pi s_n r_n(q)\right). \tag{10.4}$$

The bracket has strictly negative imaginary part in the open interval and therefore cannot vanish. □

Theorem 10.3 (RVM is the unique positive-sweep constant projective state in the one-arch family)**.** *For*

$$w_s(q) = A\sin(\pi q)e^{-i\pi s q}, \qquad s > 0, \tag{10.5}$$

the projective defect $\mathcal{P}_s = \{w_s, q\} - 2\pi^2$ is

$$\mathcal{P}_s(q) = -\frac{\pi^2(s^2-1)}{4}\,\frac{s^2\cos(2\pi q) - s^2 - 4is\sin(2\pi q) + 3\cos(2\pi q) + 9}{s^2\sin^2(\pi q) + is\sin(2\pi q) - \cos^2(\pi q)}. \tag{10.6}$$

Consequently

$$\mathcal{P}_s \equiv 0 \quad \Longleftrightarrow \quad s = 1. \tag{10.7}$$

Proof. Direct differentiation of (10.5) gives (10.6). The factor s^2-1 proves one direction. Conversely, at $q = 0$,

$$\mathcal{P}_s(0) = 3\pi^2(s^2 - 1), \tag{10.8}$$

so for $s > 0$ an identically vanishing defect forces $s = 1$. □

Thus the same value $s = 1$ is characterized in two independent geometric languages: the direct one-arch lobe is an exact circle by Theorem 4.1, and its Schwarzian is exactly the constant $2\pi^2$ by Theorem 10.3.

Proposition 10.4 (Geometric decomposition of the Schwarzian)**.** *Write*

$$w_n'(q) = v(q)e^{i\tau(q)}, \qquad v(q) > 0, \tag{10.9}$$

and let κ be the Euclidean curvature with respect to arc length s_{arc}. Then

$$\mathrm{Re}\{w_n, q\} = (\log v)'' - \frac{1}{2}(\log v)'^2 + \frac{1}{2}\tau'^2, \tag{10.10}$$

$$\mathrm{Im}\{w_n, q\} = \tau'' - (\log v)'\tau' = v^2\frac{\mathrm{d}\kappa}{\mathrm{d}s_{\mathrm{arc}}}. \tag{10.11}$$

Proof. Set $A = w_n''/w_n' = (\log v)' + i\tau'$. Since

$$\{w_n, q\} = A' - \frac{1}{2}A^2, \tag{10.12}$$

the first two expressions follow by taking real and imaginary parts. Using $\tau' = \kappa v$ converts the imaginary part to the final term in (10.11). □

10.1. **Schwarzian equation and closure moment.** Fix the continuous square-root branch determined by $w_n'(0) > 0$, and define

$$Y_n(q) = \frac{w_n(q)}{\sqrt{w_n'(q)w_n'(0)}}. \tag{10.13}$$

Then

$$Y_n(0) = 0, \qquad Y_n'(0) = 1. \tag{10.14}$$

Theorem 10.5 (Exact projective ODE)**.** *The function Y_n satisfies*

$$Y_n'' + \left(\pi^2 + \frac{1}{2}\mathcal{P}_n(q)\right) Y_n = 0. \tag{10.15}$$

Proof. Differentiating $w_n/\sqrt{w_n'}$ gives the classical identity

$$\left(\frac{w_n}{\sqrt{w_n'}}\right)'' + \frac{1}{2}\{w_n, q\}\frac{w_n}{\sqrt{w_n'}} = 0. \tag{10.16}$$

The normalization by the constant $\sqrt{w_n'(0)}$ and definition (10.2) give (10.15). □

Variation of constants around the RVM free arch gives

$$Y_n(q) = \frac{\sin(\pi q)}{\pi} - \frac{1}{2\pi}\int_0^q \sin\big(\pi(q-x)\big)\mathcal{P}_n(x)Y_n(x)\,\mathrm{d}x. \tag{10.17}$$

Because $w_n(1) = 0$, also $Y_n(1) = 0$, and (10.17) at $q = 1$ yields the exact closure orthogonality

$$\int_0^1 \mathcal{P}_n(x)\sin(\pi x)Y_n(x)\,\mathrm{d}x = 0. \tag{10.18}$$

10.2. **Correct endpoint derivative and branch.** The endpoint derivative is the place where the reciprocal error must be excluded.

Proposition 10.6 (Correct $Y_n'(1)$ quotient)**.** *With the normalization* (10.13)*,*

$$\big(Y_n'(1)\big)^2 = \frac{w_n'(1)}{w_n'(0)} = \mathcal{H}_n, \tag{10.19}$$

not the reciprocal quotient.

Proof. At $q = 1$ the terms proportional to $w_n(1)$ vanish after differentiating (10.13), so

$$Y_n'(1) = \frac{w_n'(1)}{\sqrt{w_n'(1)w_n'(0)}}. \tag{10.20}$$

Squaring gives (10.19). □

The square root is fixed by the lobe geometry rather than by an arbitrary principal-branch convention. The factor

$$D_n(q) = r_n'(q) - i\pi s_n r_n(q) \tag{10.21}$$

lies in the open lower half-plane for $0 < q < 1$, begins on the positive real axis, and ends on the negative real axis. Its continuous argument therefore changes by $-\pi$. Together with $e^{-i\pi s_n q}$ in (10.4),

$$\Delta \arg w_n' = -\pi(s_n + 1). \tag{10.22}$$

Hence the continuously transported square root gives

$$Y_n'(1) = -\sqrt{\frac{a_{n+1}}{a_n}}e^{-\frac{i}{2}(\Omega_n-\pi)}. \tag{10.23}$$

Define

$$\lambda_n = -Y_n'(1) = \sqrt{\frac{a_{n+1}}{a_n}}e^{-\frac{i}{2}(\Omega_n-\pi)}. \tag{10.24}$$

Then

$$\lambda_n^2 = \mathcal{H}_n. \tag{10.25}$$

For the exact RVM lobe, $a_{n+1} = a_n$ and $s_n = 1$, so $Y_n'(1) = -1$ and $\lambda_n = 1$.

10.3. **Correct complex moment and telescoping law.** Differentiating (10.17) and setting $q = 1$ gives

$$Y_n'(1) = -1 + \frac{1}{2}\int_0^1 \cos(\pi x)\mathcal{P}_n(x)Y_n(x)\,\mathrm{d}x. \tag{10.26}$$

Using (10.18) and (10.24), define

$$\mathcal{I}_n := \int_0^1 \mathcal{P}_n(x)e^{i\pi x}Y_n(x)\,\mathrm{d}x. \tag{10.27}$$

Then

$$\mathcal{I}_n = 2(1-\lambda_n). \tag{10.28}$$

By (9.9) and the geometrically fixed square-root branch,

$$\lambda_n = e^{(\overline{Q}_n-\overline{Q}_{n+1})/2}. \tag{10.29}$$

Therefore, for every $k \geq 1$ for which the lobe chain satisfies the stated simplicity assumptions,

$$\prod_{j=n}^{n+k-1}\left(1-\frac{\mathcal{I}_j}{2}\right) = \exp\left(\frac{\overline{Q}_n-\overline{Q}_{n+k}}{2}\right). \tag{10.30}$$

Remark 10.7 (What the Schwarzian block does and does not add). Equations (10.18), (10.28), and (10.30) are exact. They reconstruct endpoint scattering from an interior projective defect. At present they provide no sign, sector, monotonicity, or norm constraint on $\mathcal{I}_n$ that would imply (1.1). The projective block is therefore a structural identity layer, not the missing confinement theorem.

11. Euclidean arc length and phase-coordinate geometry

The Euclidean length of the nth lobe is

$$L_n = \int_{\gamma_n}^{\gamma_{n+1}} \sqrt{1+\left|\zeta'\left(\frac{1}{2}+it\right)\right|^2}\,\mathrm{d}t. \tag{11.1}$$

Using (3.6),

$$L_n = \int_{\gamma_n}^{\gamma_{n+1}} \sqrt{1+Z'(t)^2+\vartheta'(t)^2Z(t)^2}\,\mathrm{d}t. \tag{11.2}$$

Define

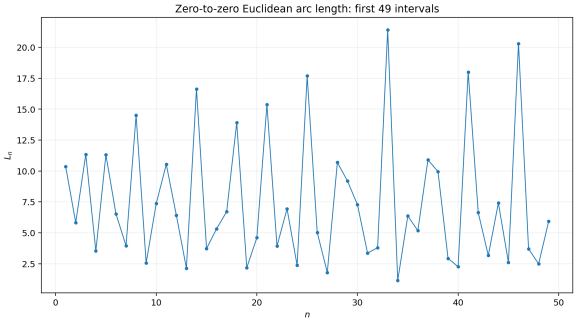
$$R_n = \frac{L_n}{g_n}. \tag{11.3}$$

Let

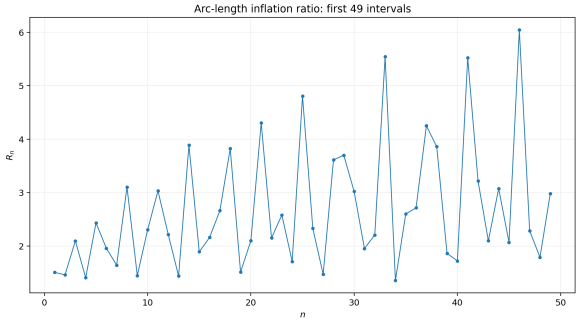
$$q_\vartheta(t) = \frac{\vartheta(t)}{\pi}. \tag{11.4}$$

Then

$$\mathrm{d}q_\vartheta = \frac{\vartheta'(t)}{\pi}\,\mathrm{d}t, \tag{11.5}$$



(A) Euclidean lobe lengths L_n.



(B) Arc inflation ratios R_n.

FIGURE 7. Zero-to-zero Euclidean arc geometry for the carried first-49 computation.

and, writing $r(q_\vartheta) = |Z(t(q_\vartheta))|$,

$$L_n = \int_{q_n}^{q_{n+1}} \sqrt{\left(\frac{\pi}{\vartheta'(t(q_\vartheta))}\right)^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}q_\vartheta}\right)^2 + \pi^2 r(q_\vartheta)^2}\,\mathrm{d}q_\vartheta. \tag{11.6}$$

The integration-domain length is exactly

$$q_{n+1} - q_n = s_n. \tag{11.7}$$

Thus the local spacing variable is the phase-coordinate width of the Euclidean lobe itself. Montgomery's pair-correlation problem [3] concerns a different, all-pairs statistic; the identification in (11.7) is specifically a nearest-neighbor geometric statement.

For the one-arch approximation

$$Z(t) \approx A_n \sin\left(\pi \frac{t-\gamma_n}{g_n}\right) \tag{11.8}$$

and locally frozen ϑ', set

$$a_n^{\mathrm{arc}} = \frac{\pi A_n}{g_n}, \qquad b_n^{\mathrm{arc}} = A_n \vartheta'(m_n), \qquad m_n = \frac{\gamma_n + \gamma_{n+1}}{2}. \tag{11.9}$$

Then

$$R_n^{\mathrm{arch}} = \frac{2}{\pi}\sqrt{1+(a_n^{\mathrm{arc}})^2}\,E\left(\frac{(a_n^{\mathrm{arc}})^2 - (b_n^{\mathrm{arc}})^2}{1+(a_n^{\mathrm{arc}})^2}\right), \tag{11.10}$$

where E is the complete elliptic integral of the second kind.

12. Eccentricity observables carried from the preceding computation

For a rotated lobe in uniform phase coordinate define

$$\mathcal{C}_n = \int_0^1 \widehat{W}_n(q)\,\mathrm{d}q, \tag{12.1}$$

$$E_n = \frac{2|\mathcal{C}_n|}{A_n}, \tag{12.2}$$

and

$$D_n = \frac{1}{A_n}\left[\int_0^1 \left(|\widehat{W}_n(q) - \mathcal{C}_n| - |\mathcal{C}_n|\right)^2 \mathrm{d}q\right]^{1/2}. \tag{12.3}$$

For the one-arch phase model,

$$\frac{\mathcal{C}_n^{\mathrm{arch}}}{A_n} = \frac{1+e^{-i\pi s_n}}{\pi(1-s_n^2)}, \tag{12.4}$$

with continuous value $-i/2$ at $s_n = 1$, and therefore

$$E_n^{\mathrm{arch}} = \frac{4|\cos(\pi s_n/2)|}{\pi|1-s_n^2|}. \tag{12.5}$$

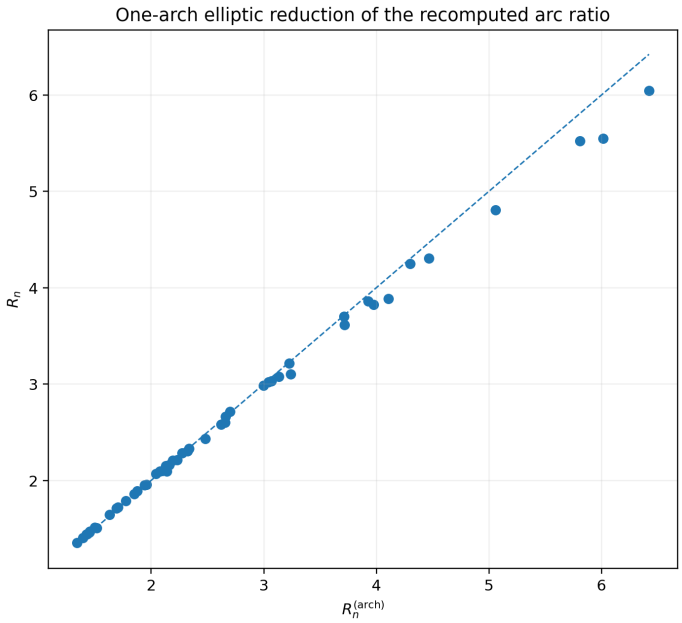
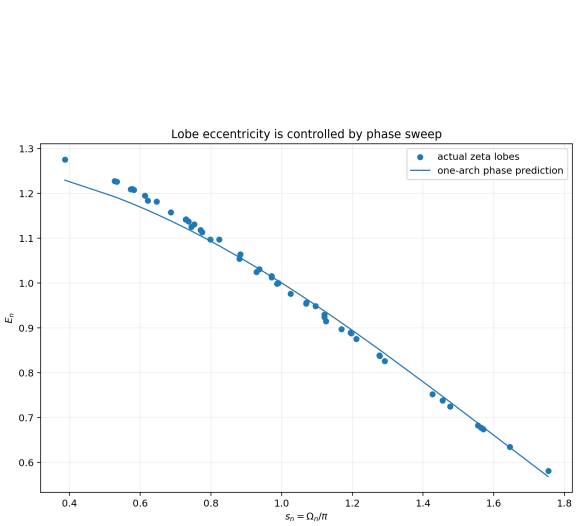
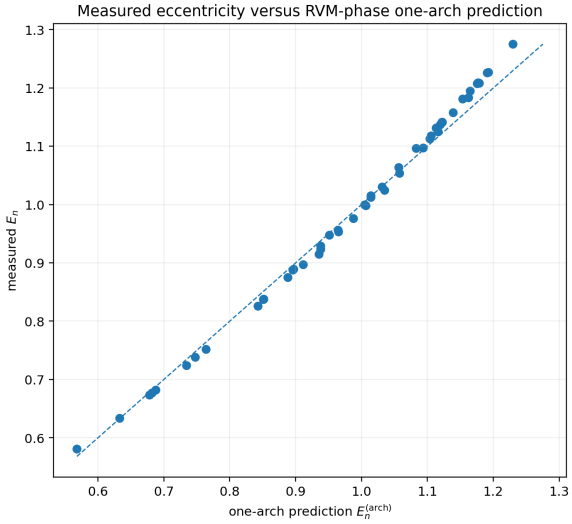


FIGURE 8. Measured arc inflation versus the one-arch elliptic reduction.



(A) Eccentricity versus phase sweep.



(B) Measured versus one-arch eccentricity.

FIGURE 9. The carried numerical eccentricity diagnostics.

The carried first-49 values give

$$\mathrm{corr}(E_n, E_n^{\mathrm{arch}}) = 0.997589, \qquad \mathrm{RMSE} = 0.017527, \qquad \mathrm{MAE} = 0.014155, \tag{12.6}$$

and the carried arc computation gives

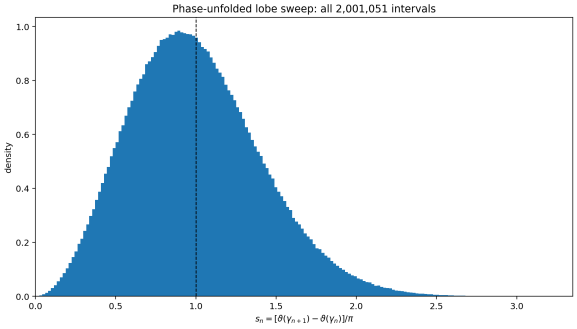
$$\mathrm{corr}(R_n, R_n^{\mathrm{arch}}) = 0.999162, \qquad \mathrm{RMSE} = 0.116004. \tag{12.7}$$

These are numerical statements, not analytic hypotheses used in the exact theorems above.

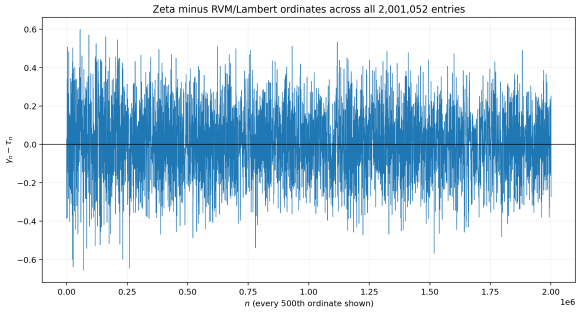
13. Full carried phase/RVM audit

The package carries the previously verified 2,001,052 zeta ordinates and the same number of RVM/Lambert ordinates. Version 5 does not inspect or recompute those binary files. The preceding audit reported

$$N = 2{,}001{,}052, \qquad N_{\mathrm{intervals}} = 2{,}001{,}051, \tag{13.1}$$



(A) Phase-unfolded sweep distribution.



(B) $\gamma_n - \tau_n$ over the complete carried corpus.

FIGURE 10. Global numerical diagnostics carried unchanged into Version 5.

$$\mathrm{RMS}(\gamma_n - \tau_n) = 0.188066300, \qquad \max_n |\gamma_n - \tau_n| = 0.993710851, \tag{13.2}$$

and

$$\mathrm{mean}(s_n) = 0.999999964, \qquad \mathrm{sd}(s_n) = 0.407523227. \tag{13.3}$$

14. What remains genuinely open in this construction

The exact RVM theorem, Theorem 6.1, exhibits a complete no-exit mechanism for the tangent-circle skeleton. The exact zeta lobes possess the same global quarter-turn equation (5.2), but the deformation from the RVM skeleton is not currently controlled by an invariant strong enough to imply (1.1).

The reciprocal chart does not alter the logical event by Proposition 7.1. The projective defect does not close the gap either: Remark 10.7 records that the Schwarzian moment identities recover endpoint holonomy but do not yet constrain it in a zero-excluding region.

Accordingly, the unresolved analytic target can be stated without introducing a new classical RH criterion:

(14.1) derive from the exact lobe geometry and $F_\delta = -iF_t$ a global invariant that excludes $F = 0$ for $\delta \neq 0$.

The target is first-order in the evolution variable; the missing ingredient is a global rigidity inequality or invariant of the specific zeta lobe family.

15. Provenance of the construction

The zeta function, Hardy's Z-function, the Riemann–Siegel phase, the smooth Riemann–von Mangoldt phase, Lambert W, Cauchy–Riemann equations, Schwarzian derivatives, and variation of constants are classical ingredients. Standard background on the zeta function is available in [1, 2]; the Lambert W reference used for the smooth inverse-counting parametrization is [4].

The construction developed here keeps the three-dimensional trace (3.2) as the primary object and organizes it into zero-to-zero lobes carrying Euclidean length, phase sweep, eccentricity, reciprocal scattering, and transverse quarter-turn dynamics. The specific combination of the tangent-circle RVM skeleton, reciprocal straightening, endpoint-normalized lobe factorization, corrected scattering cocycle, and Schwarzian projective defect belongs to this program.

The two corrections incorporated in Version 5 are essential: the endpoint derivative identity is

$$\big(Y_n'(1)\big)^2 = \frac{w_n'(1)}{w_n'(0)}, \tag{15.1}$$

not its reciprocal, and the scattering potential is

$$Q_n = -\log a_n - i\psi_n, \tag{15.2}$$

which gives the modulus a_{n+1}/a_n in the correct direction.

16. Conclusion

The Violet Thread is naturally described as a chain of eccentric zero-to-zero lobes rather than a helix centered on the zero axis. A unit phase sweep combined with a one-arch radial profile gives an exact tangent circle. The smooth RVM/Lambert construction realizes that geometry exactly and, after transverse continuation, has the explicit no-exit barrier (6.4).

In reciprocal coordinates the same model becomes a straight scattering branch. This change of chart does not create a distinct zero problem: Proposition 7.1 identifies the reciprocal pole and the direct zero as the same event. Its genuine role is to simplify the model geometry and to expose endpoint scattering quantities.

For exact zeta lobes, the corrected holonomy

$$\mathcal{H}_n = \frac{a_{n+1}}{a_n} e^{-i(\Omega_n - \pi)} \tag{16.1}$$

and projective defect $\mathcal{P}_n$ produce the exact chain

$$\mathcal{P}_n \longrightarrow Y_n \longrightarrow \mathcal{I}_n \longrightarrow \lambda_n \longrightarrow \mathcal{H}_n \longrightarrow \overline{Q}_n - \overline{Q}_{n+1}. \tag{16.2}$$

The chain is mathematically consistent after the reciprocal/sign corrections, but it is descriptive until an additional rigidity inequality is proved. The analytic target therefore remains sharply defined inside the same construction: transfer the RVM no-exit mechanism to the exact lobe family using an invariant of the global first-order transverse flow.

Appendix A. Carried first-49 Euclidean arc table

n	γ_n	γ_{n+1}	g_n	L_n	R_n	s_n
1	14.134725142	21.022039639	6.887314	10.366625	1.505177	1.120464
2	21.022039639	25.010857580	3.988818	5.821954	1.459569	0.823387
3	25.010857580	30.424876126	5.414019	11.347658	2.095977	1.277467
4	30.424876126	32.935061588	2.510185	3.527308	1.405198	0.646207
5	32.935061588	37.586178159	4.651117	11.313733	2.432477	1.276284
6	37.586178159	40.918719012	3.332541	6.525882	1.958230	0.971574
7	40.918719012	43.327073281	2.408354	3.954497	1.641991	0.729252
8	43.327073281	48.005150881	4.678078	14.508230	3.101323	1.476437
9	48.005150881	49.773832478	1.768682	2.558463	1.446537	0.577518
10	49.773832478	52.970321478	3.196489	7.361974	2.303144	1.068877
11	52.970321478	56.446247697	3.475926	10.542035	3.032871	1.197124
12	56.446247697	59.347044003	2.900796	6.417297	2.212254	1.025228
13	59.347044003	60.831778525	1.484735	2.134053	1.437330	0.533553
14	60.831778525	65.112544048	4.280766	16.638613	3.886831	1.570144
15	65.112544048	67.079810529	1.967266	3.718671	1.890273	0.736783
16	67.079810529	69.546401711	2.466591	5.326509	2.159462	0.936736
17	69.546401711	72.067157674	2.520756	6.712577	2.662922	0.971691
18	72.067157674	75.704690699	3.637533	13.914973	3.825387	1.426798
19	75.704690699	77.144840069	1.440149	2.176283	1.511151	0.572652
20	77.144840069	79.337375020	2.192535	4.602265	2.099061	0.880017
21	79.337375020	82.910380854	3.573006	15.382321	4.305148	1.454644
22	82.910380854	84.735492981	1.825112	3.922717	2.149302	0.752565
23	84.735492981	87.425274613	2.689782	6.941343	2.580634	1.120470
24	87.425274613	88.809111208	1.383837	2.364722	1.708816	0.581616
25	88.809111208	92.491899271	3.682788	17.699808	4.806089	1.564427
26	92.491899271	94.651344041	2.159445	5.028165	2.328453	0.928236
27	94.651344041	95.870634228	1.219290	1.792394	1.470031	0.527586
28	95.870634228	98.831194218	2.960560	10.692946	3.611799	1.291244
29	98.831194218	101.317851006	2.486657	9.200234	3.699841	1.095477
30	101.317851006	103.725538040	2.407687	7.271512	3.020123	1.069947
31	103.725538040	105.446623052	1.721085	3.359732	1.952101	0.770293
32	105.446623052	107.168611184	1.721988	3.798904	2.206115	0.775172
33	107.168611184	111.029535543	3.860924	21.415034	5.546608	1.753940

n	γ_n	γ_{n+1}	g_n	L_n	R_n	s_n
34	111.029535543	111.874659177	0.845124	1.144932	1.354751	0.386799
35	111.874659177	114.320220915	2.445562	6.361304	2.601163	1.124991
36	114.320220915	116.226680321	1.906459	5.173808	2.713831	0.882782
37	116.226680321	118.790782866	2.564103	10.900164	4.251064	1.195136
38	118.790782866	121.370125002	2.579342	9.953938	3.859099	1.211127
39	121.370125002	122.946829294	1.576704	2.933196	1.860334	0.744648
40	122.946829294	124.256818554	1.309989	2.254766	1.721209	0.621133
41	124.256818554	127.516683880	3.259865	18.008263	5.524235	1.555161
42	127.516683880	129.578704200	2.062020	6.634166	3.217314	0.990584
43	129.578704200	131.087688531	1.508984	3.163710	2.096582	0.728222
44	131.087688531	133.497737203	2.410049	7.413230	3.075967	1.168789
45	133.497737203	134.756509753	1.258773	2.604796	2.069315	0.613221
46	134.756509753	138.116042055	3.359532	20.311522	6.045937	1.645739
47	138.116042055	139.736208952	1.620167	3.702662	2.285358	0.798342
48	139.736208952	141.123707404	1.387498	2.480013	1.787399	0.686072
49	141.123707404	143.111845808	1.988138	5.929834	2.982606	0.986847

Appendix B. Carried first-49 eccentric-lobe table

n	s_n	A_n	$\mathrm{Re}\,\mathcal{C}_n$	$\mathrm{Im}\,\mathcal{C}_n$	E_n	D_n	E_n^{arch}
1	1.120464	2.340551	-0.197557	-1.069503	0.929351	0.061686	0.937571
2	0.823387	1.459266	0.213980	-0.771070	1.096730	0.094567	1.082844
3	1.277467	2.847473	-0.467973	-1.096033	0.837064	0.136510	0.850629
4	0.646207	0.925523	0.288074	-0.464831	1.181729	0.189896	1.153336
5	1.276284	2.942394	-0.569632	-1.094018	0.838387	0.136454	0.851303
6	0.971574	1.812834	0.081551	-0.914154	1.012541	0.017313	1.014081
7	0.729252	1.129621	0.254384	-0.592648	1.141863	0.145029	1.122018
8	1.476437	3.664837	-0.863191	-1.008247	0.724330	0.221013	0.734307
9	0.577518	0.713345	0.269293	-0.337016	1.209490	0.225516	1.176783
10	1.068877	2.061177	-0.181974	-0.965993	0.953808	0.038440	0.964823
11	1.197124	2.913512	-0.382057	-1.235552	0.887776	0.100247	0.895805
12	1.025228	1.816095	0.036943	-0.885544	0.976066	0.020841	0.987285
13	0.533553	0.603289	0.243649	-0.278203	1.225990	0.247800	1.190566
14	1.570144	4.166936	-1.145451	-0.811102	0.673659	0.256740	0.678236
15	0.736783	1.134191	0.276452	-0.582461	1.136911	0.140711	1.119024
16	0.936736	1.551068	0.058545	-0.797230	1.030743	0.033477	1.030967
17	0.971691	2.004530	0.001540	-1.017693	1.015394	0.017383	1.014024
18	1.426798	3.641859	-0.733955	-1.155689	0.751843	0.201781	0.763769
19	0.572652	0.632316	0.243141	-0.294845	1.208782	0.227034	1.178356
20	0.880017	1.368228	0.081166	-0.716356	1.053829	0.063916	1.057534
21	1.454644	3.998567	-0.963358	-1.117642	0.738028	0.215416	0.747269
22	0.752565	1.238064	0.280679	-0.641532	1.131195	0.132691	1.112669
23	1.120470	1.952783	-0.142953	-0.890642	0.923852	0.061227	0.937568
24	0.581616	0.740985	0.267355	-0.358679	1.207469	0.223290	1.175449
25	1.564427	4.477141	-1.231837	-0.883265	0.677118	0.257169	0.681668
26	0.928236	1.505698	0.160950	-0.754544	1.024799	0.041524	1.035022
27	0.527586	0.519217	0.212790	-0.237095	1.227156	0.250352	1.192361
28	1.291244	2.933506	-0.671815	-1.008007	0.825884	0.142695	0.842757
29	1.095477	2.698327	-0.166322	-1.268317	0.948125	0.049823	0.950863
30	1.069947	2.126243	-0.049191	-1.015659	0.956475	0.038085	0.964266
31	0.770293	1.023117	0.211215	-0.531506	1.118027	0.121926	1.105401
32	0.775172	1.198408	0.196300	-0.637329	1.112934	0.119259	1.103377
33	1.753940	5.193290	-1.310908	-0.745479	0.580769	0.316608	0.567986
34	0.386799	0.322351	0.169868	-0.115712	1.275223	0.318951	1.229286
35	1.124991	1.800191	-0.233571	-0.789519	0.914730	0.064462	0.935145
36	0.882782	1.613611	0.134317	-0.848048	1.064221	0.062578	1.056266
37	1.195136	3.102800	-0.465731	-1.298326	0.889088	0.099593	0.896905
38	1.211127	2.814119	-0.287478	-1.197322	0.875123	0.107082	0.888027
39	0.744648	0.889614	0.215212	-0.451735	1.124938	0.134423	1.115870
40	0.621133	0.697945	0.218616	-0.350456	1.183625	0.200187	1.162160
41	1.555161	4.592177	-1.304854	-0.865631	0.681975	0.254118	0.687228
42	0.990584	2.022093	0.093840	-1.006610	0.999929	0.016746	1.004694
43	0.728222	1.004221	0.238322	-0.521409	1.141767	0.145333	1.122426

n	s_n	A_n	$\mathrm{Re}\,\mathcal{C}_n$	$\mathrm{Im}\,\mathcal{C}_n$	E_n	D_n	E_n^{arch}
44	1.168789	2.097730	-0.245463	-0.908151	0.896912	0.084512	0.911408
45	0.613221	0.869925	0.288251	-0.432411	1.194770	0.206969	1.164882
46	1.645739	5.050324	-1.375512	-0.819209	0.634011	0.287141	0.632816
47	0.798342	1.168116	0.240775	-0.593840	1.097143	0.106380	1.093630
48	0.686072	0.758449	0.203407	-0.389108	1.157800	0.166754	1.138697
49	0.986847	1.799303	-0.062979	-0.895959	0.998353	0.019753	1.006548

Appendix C. Reproducibility package

The archive contains one principal TeX source, its compiled PDF, the fifteen PNG files actually used by the manuscript, the consolidated Python analysis script from the preceding verified computation, the carried CSV/TeX outputs, and the two binary ordinate files. The binary files are copied unchanged into Version 5 and are not read by the Version 5 revision process.

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